

[illegible]

Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F.Paste				Paste Mask	GBR
F.Silkscreen				Direct Printing	Legend
F.Mask		0.02mm		Solder Resist	GBR
Copper	F.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	GBR
Copper	In1.Cu (GND)	0.035mm (1.00oz)		Plane	GBR
Core		1.528mm	FR4	Dielectric	GBR
Copper	In2.Cu (Sig, PWR)	0.035mm (1.00oz)		Plane	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	GBR
Copper	B.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
B.Mask		0.02mm		Solder Resist	GBR
B.Silkscreen				Direct Printing	Legend
B.Paste				Paste Mask	GBR

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating

Dimensions:
Overall width: 100.0000 mm
Overall height: 100.0000 mm
Hole diameter: 5.0000 mm
Radius: R 3.2000 mm

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts,GBR" SUFFIX.
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SURFACE FINISH: HAL SNPB
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:
 - FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - Tg 170 C OR EQUIVALENT.
 - EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	100.000 × 100.000 mm
BOARD THICKNESS	2.278 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	1.000 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.300 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm

Lim. resistor
PFC

HMI
Ext ctrl

SWD Status

1 kV strobe
04/Jul/2025

Boot

Light sense

Cap+

Lamp trig

Lamp+

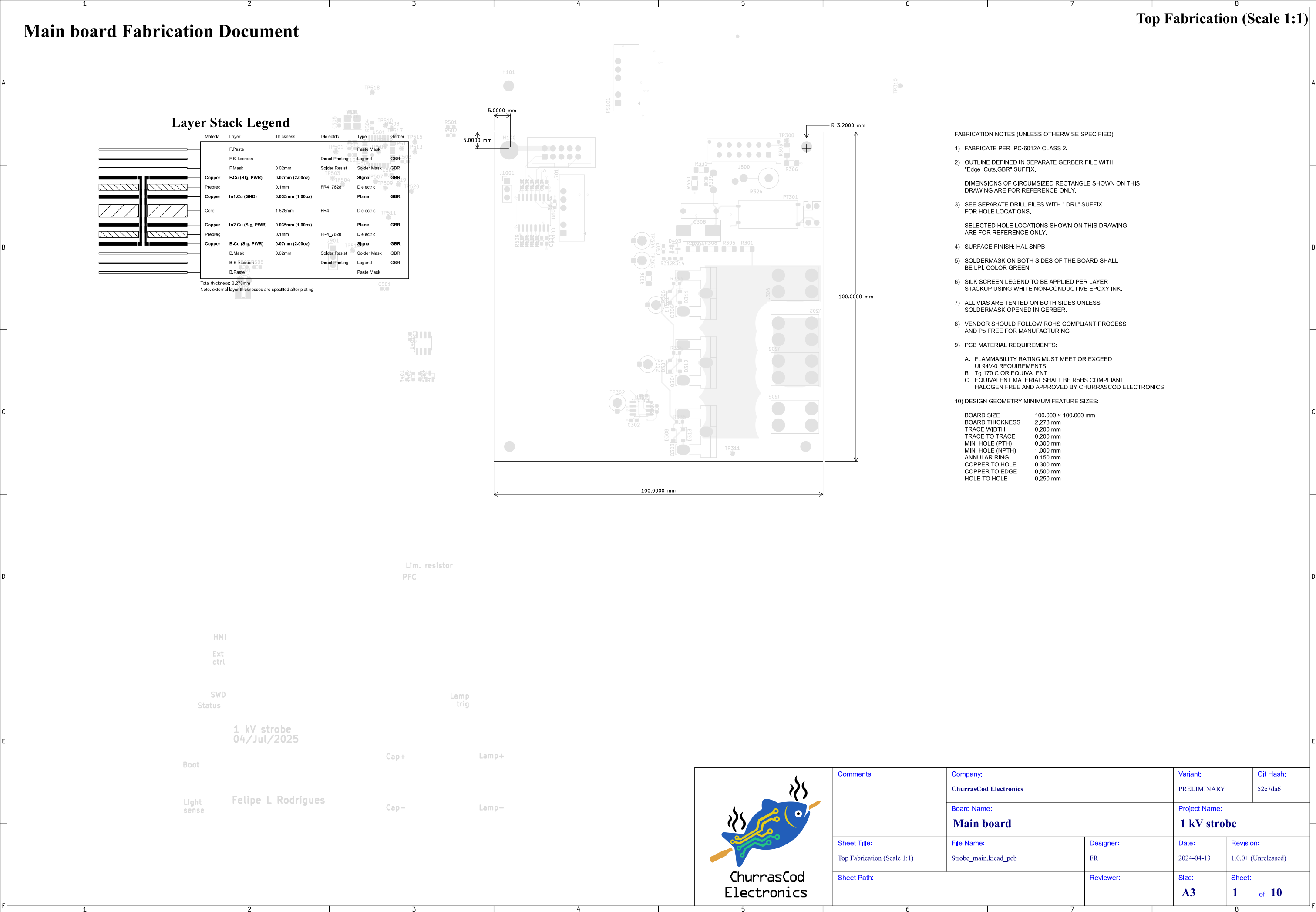
Cap-

Lamp-

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	GIt Hash: 52c7da6
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:	Reviewer:		Size: A3	Sheet: 1 of 10

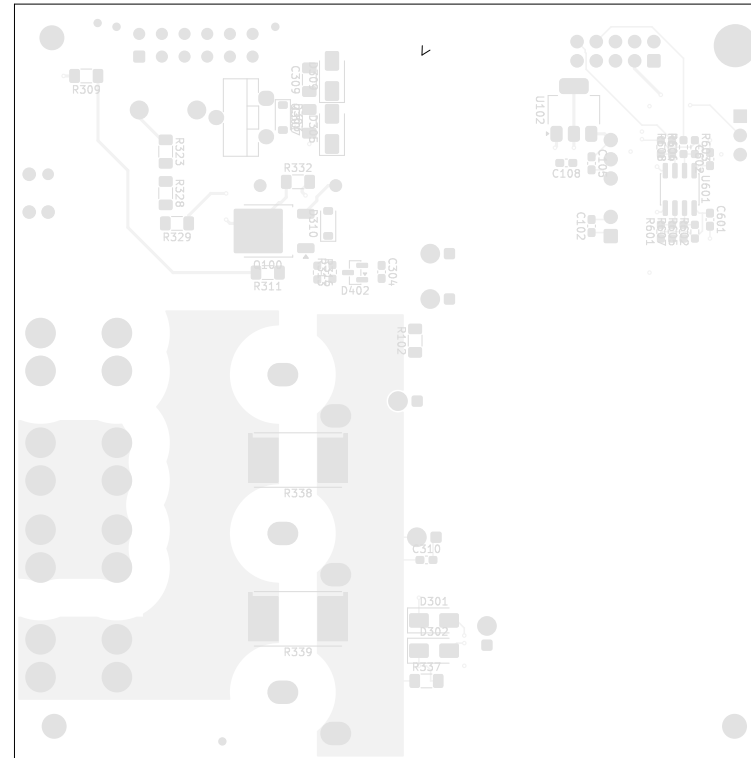
[illegible][illegible][illegible]

- [illegible]



Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



1 kV strobe
04/Jul/25
Felipe L Rodrigues



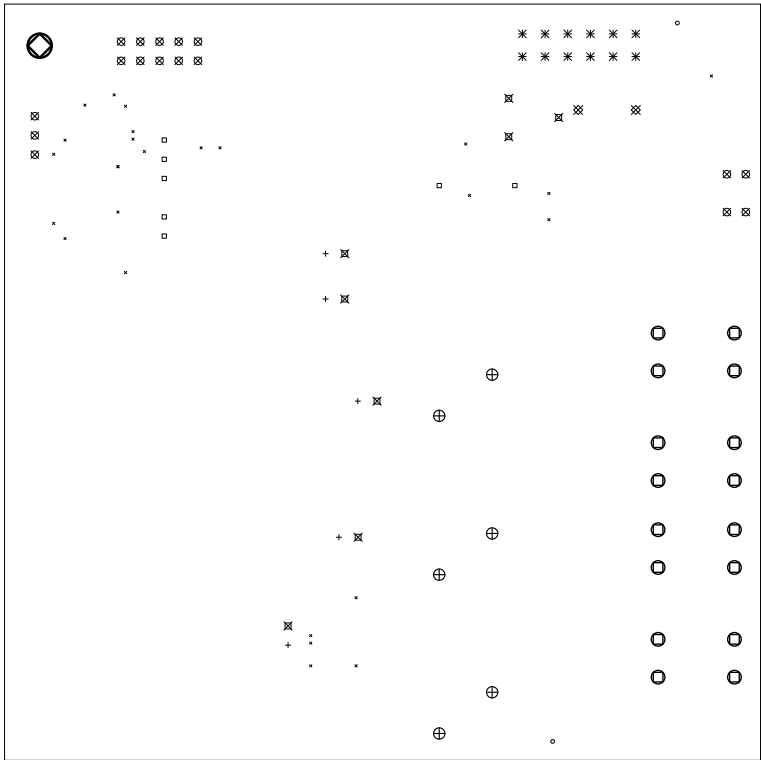
Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 52e7da6
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 2 of 10

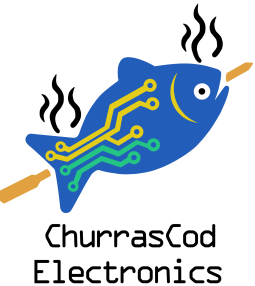
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	33	0,30mm (11,81mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Via
○	24	0,50mm (19,69mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
+	5	0,70mm (27,56mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
□	12	0,80mm (31,50mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
◇	2	0,90mm (35,43mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊗	20	1,00mm (39,37mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
※	12	1,02mm (40,16mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊠	8	1,10mm (43,31mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊞	2	1,20mm (47,24mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊕	6	1,50mm (59,06mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊖	16	1,80mm (70,87mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊗	1	3,20mm (125,98mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
Total 141						



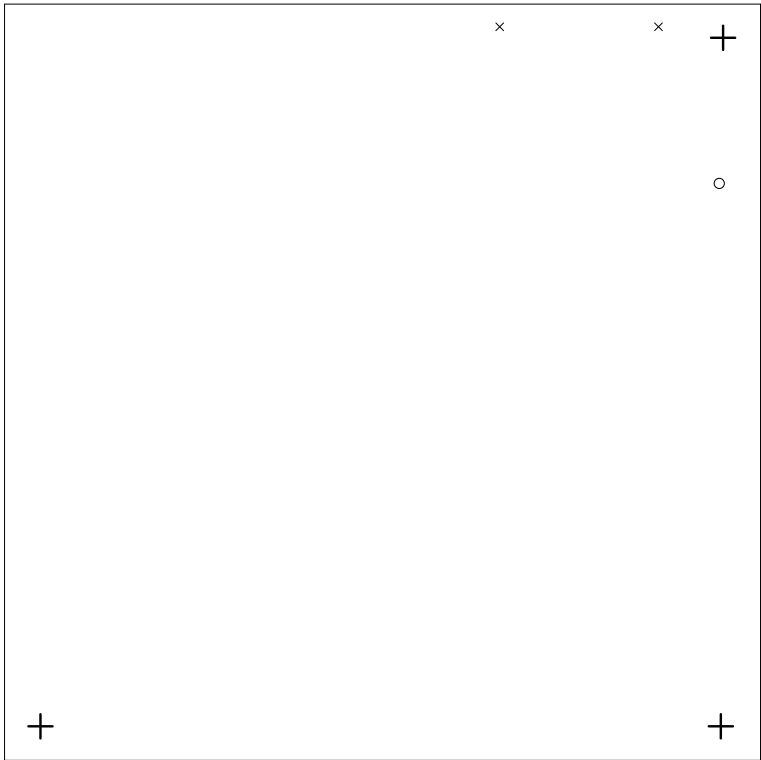
	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 52e7da6
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)	
	Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10	

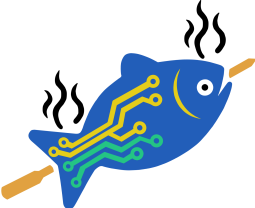
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

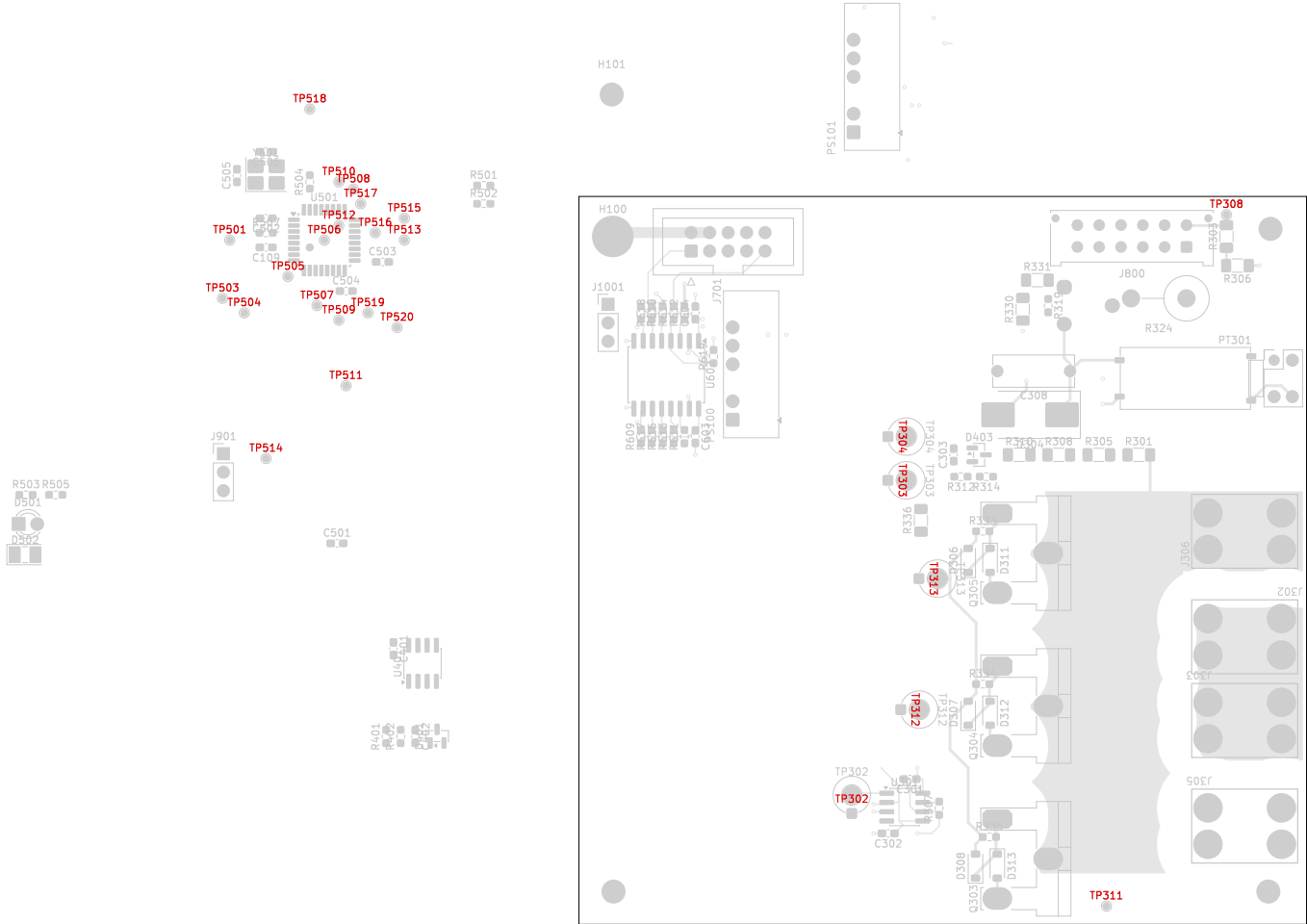
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1,02mm (40,16mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
○	1	1,35mm (53,15mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
Total 7						



 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 52e7da6
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer:	Size: A3	Sheet: 4 of 10

Main board Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP302	LMP_EN	42.50	21.50
TP303	SH0	48.73	66.00
TP304	SH0	48.73	72.00
TP308	S1k	94.00	102.50
TP310	TP_PRI	128.50	119.00
TP311	LMP	77.50	7.50
TP312	SH0	50.50	34.50
TP313	SH0	53.00	52.50
TP501	Net-(U501-PA0)	-43.00	99.00
TP503	SUP_TX	-44.00	91.00
TP504	SUP_RX	-41.00	89.00
TP505	Mon_S1k_buf	-35.00	94.00
TP506	Net-(U501-PB0)	-30.00	99.00
TP507	Mon_IL_prot	-31.00	90.00
TP508	Net-(U501-PB3)	-26.00	106.00
TP509	Mon_LMP_buf	-28.00	88.00
TP510	Net-(U501-PB4)	-28.00	107.00
TP511	Net-(U501-PA7)	-27.00	79.00
TP512	Net-(U501-PB5)	-28.00	101.00
TP513	Vbus_ctrl	-19.00	99.00
TP514	Light_sns	-38.00	69.00
TP515	INT_EN	-19.00	102.00
TP516	Net-(U501-PA12)	-23.00	100.00
TP517	Net-(U501-PA15)	-25.00	104.00
TP518	BOOT0	-32.00	117.00
TP519	SH0	-24.00	89.00
TP520	SH0	-20.00	87.00

Lim. resistor
PFC

HMI
Ext
ctrl

SWD
Status

1 kV strobe
04/Jul/2025

Boot

Light
sense

Felipe L Rodrigues

Cap+

Cap-

Lamp trig

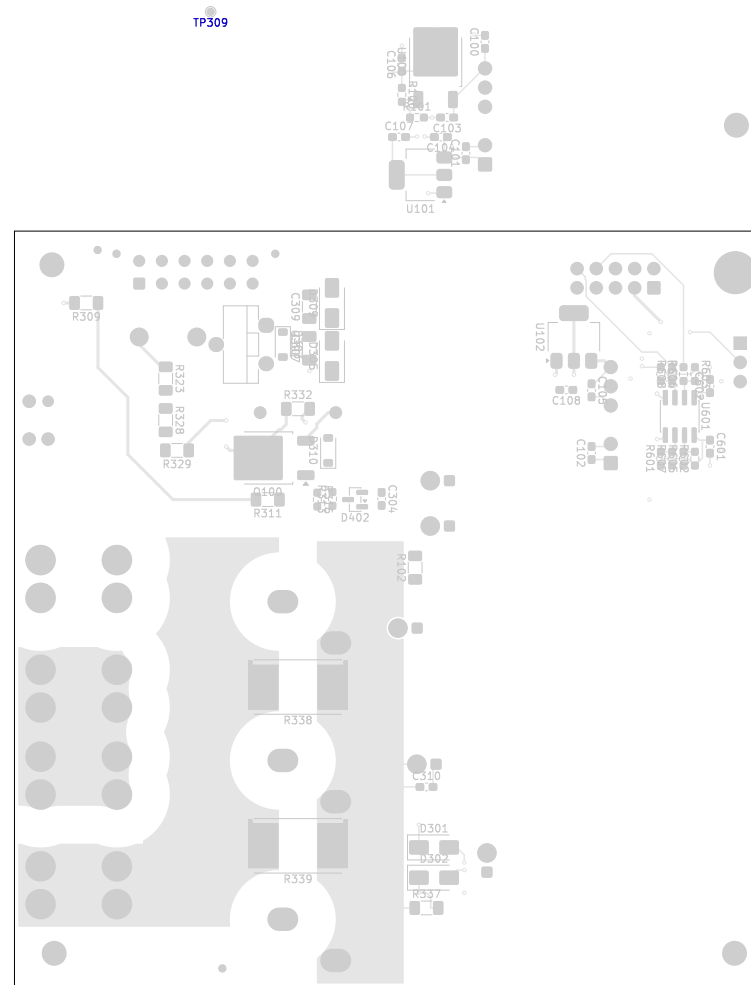
Lamp+

Lamp-

	Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 52e7da6
	Sheet Title: Top Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 5 of 10

Main board Fabrication Document

Bottom Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP309	TP_REF	79.05	134.00
TP502	TP502 Net-(U501-PA1)	-32.00	98.00

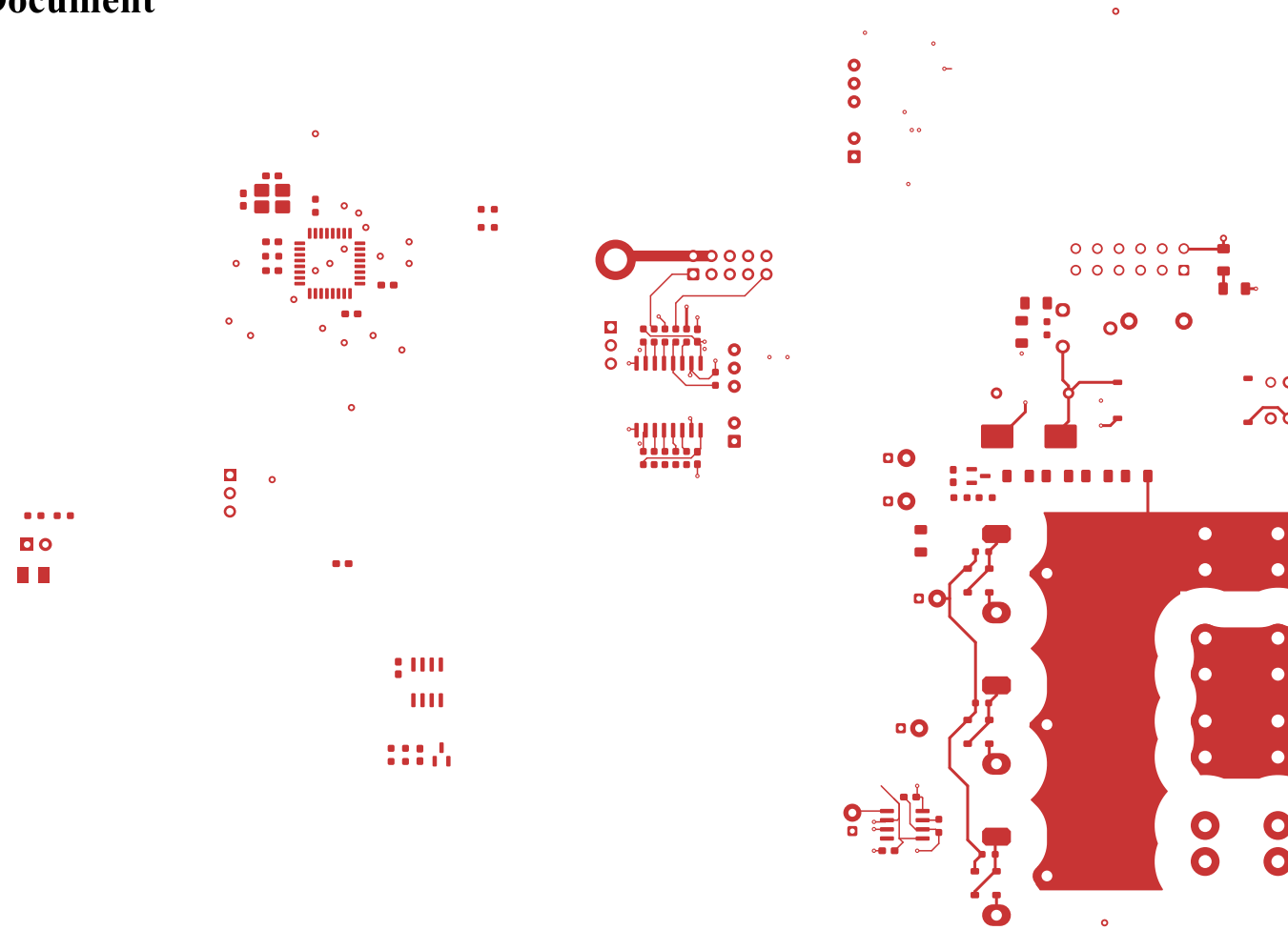
1 kV strobe
04/Jul/25
Felipe L Rodrigues



Comments:	Company: ChurrasCod Electronics	Variant: PRELIMINARY	Git Hash: 52e7da6	
	Board Name: Main board	Project Name: 1 kV strobe		
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 6 of 10

Main board Fabrication Document

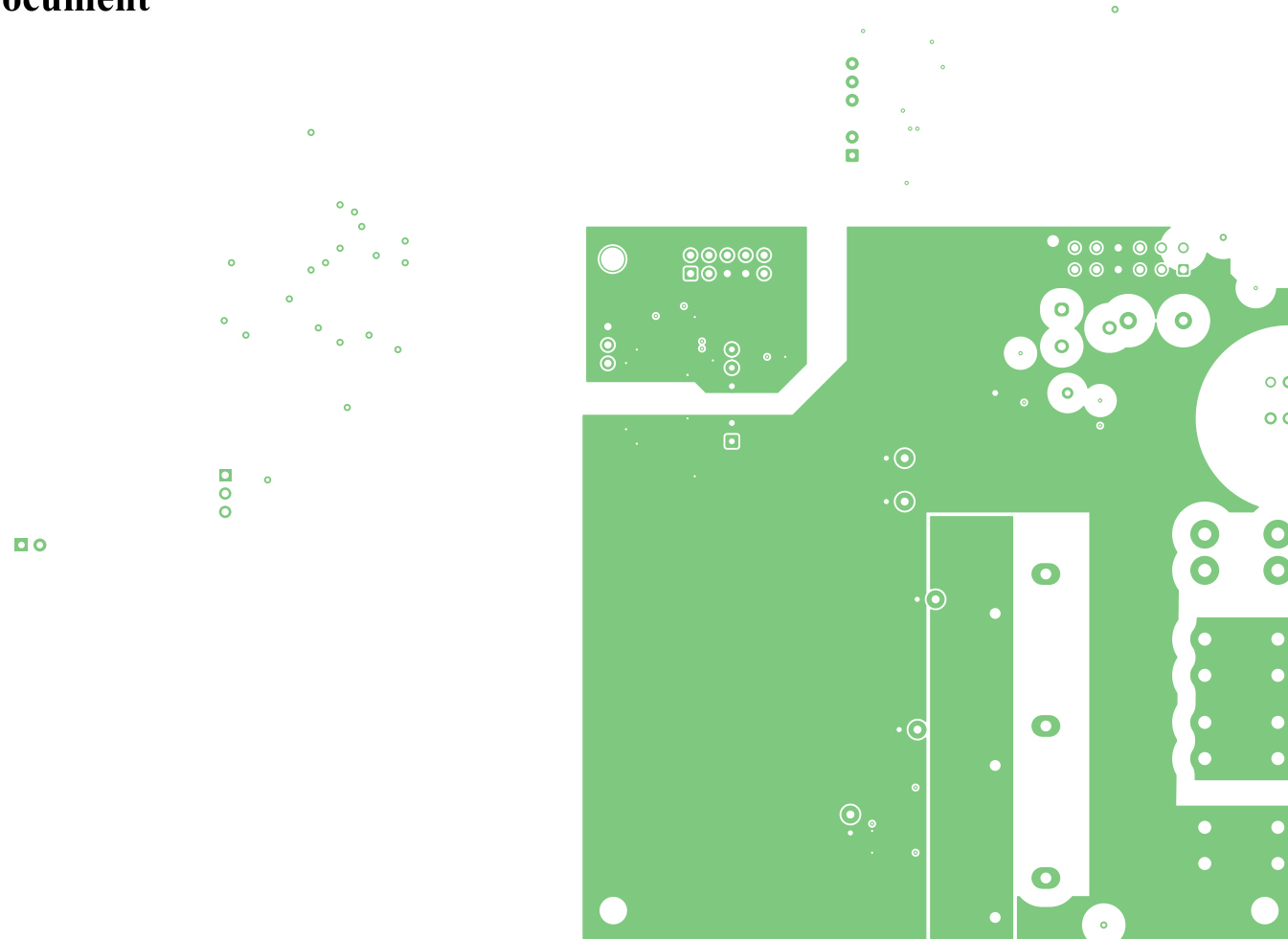
F.Cu (Sig, PWR) (Scale 1:1)



Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 52e7da6
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: F.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 7 of 10

Main board Fabrication Document

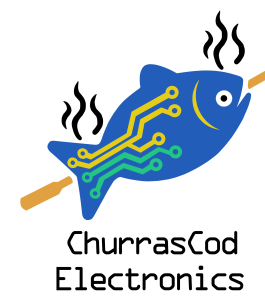
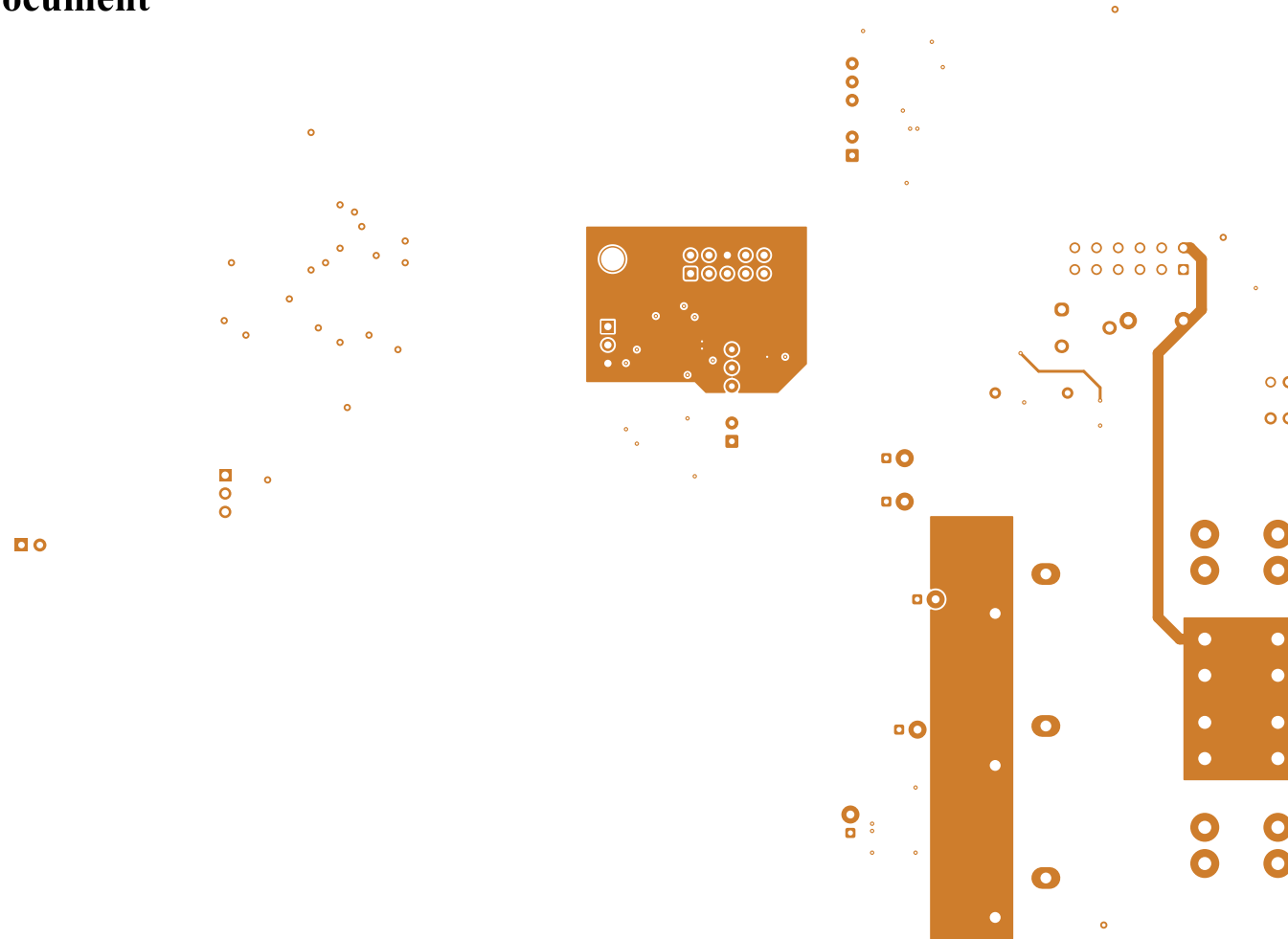
In1.Cu (GND) (Scale 1:1)	



Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 52e7da6
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: In1.Cu (GND) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 8 of 10

Main board Fabrication Document

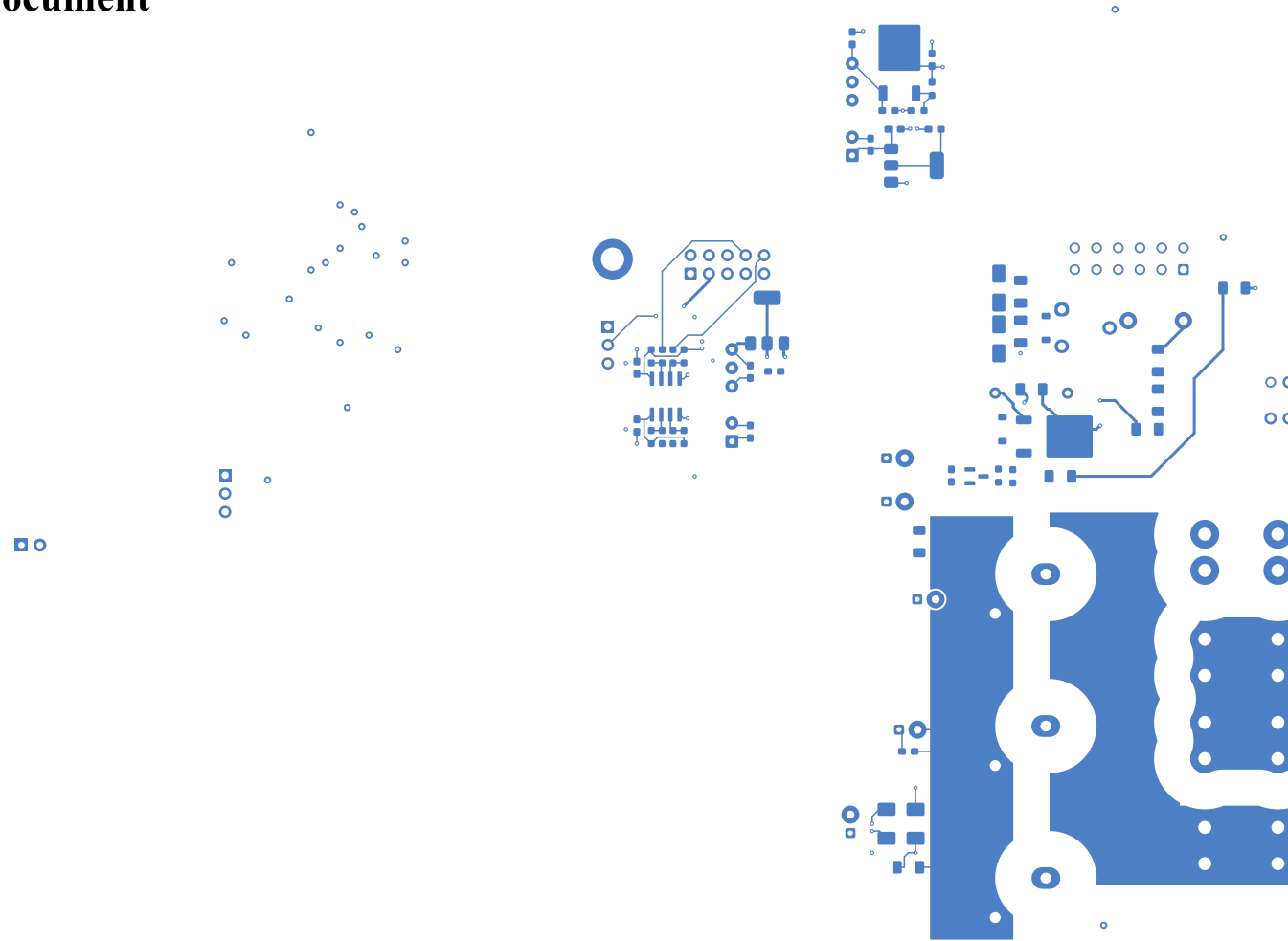
In2.Cu (Sig, PWR) (Scale 1:1)



Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 52e7da6
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 9 of 10

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 52e7da6
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 10 of 10